

Class 7: Machine Learning 1

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Background

Today we will explore some core machine learning methods that are very popular in bioinformatics. These include **clustering** and **dimensionality reduction**.

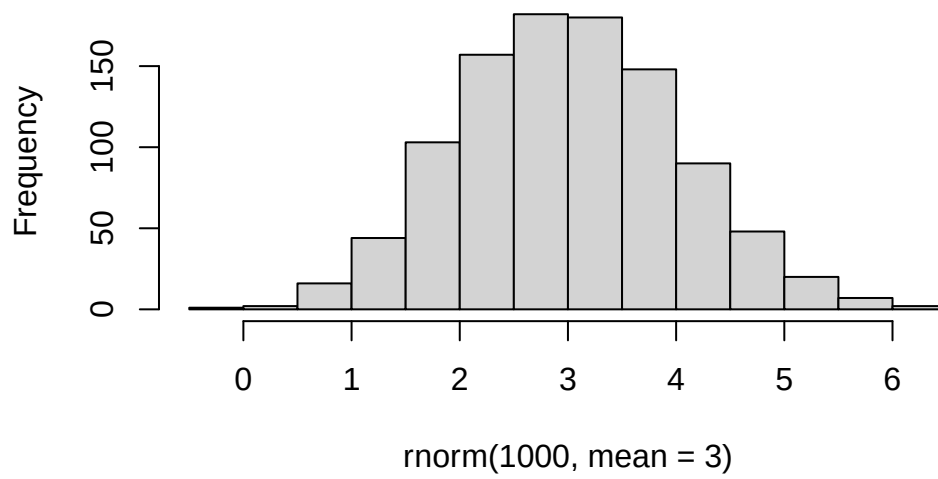
K-means clustering

The main function in “base” R for K-means clustering is called `kmeans()`

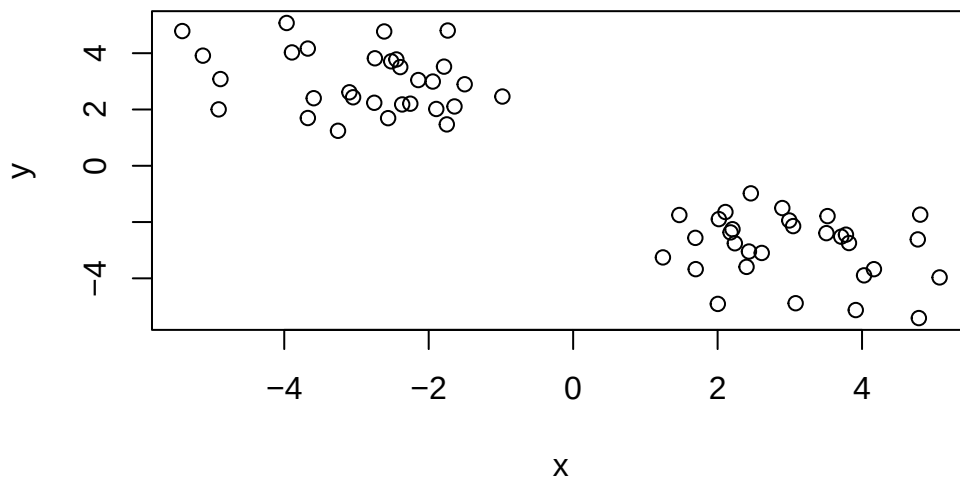
Before we go too deep, let’s make up some “simple” data that we can cluster and know if we are getting a good answer or not. To do this we can use the `rnorm()`

```
hist(rnorm(1000, mean=3))
```

Histogram of rnorm(1000, mean = 3)



```
x <- c(rnorm(30, mean=-3), rnorm(30, mean=3))  
z <- cbind(x, y=rev(x))  
plot(z)
```



Now we can run `kmeans()` on this input `z` and see what the results look like.

```
km <- kmeans(z, 2)
```

```
attributes(km)
```

```
$names
[1] "cluster"      "centers"      "totss"        "withinss"     "tot.withinss"
[6] "betweenss"    "size"         "iter"         "ifault"
$class
[1] "kmeans"
```

Q. How many points are in each cluster?

```
km$size
```

```
[1] 30 30
```

Q. What component of your result object details cluster assignment/membership?

```
km$cluster
```

```
[1] 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 1 1 1 1 1 1 1 1
[39] 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
```

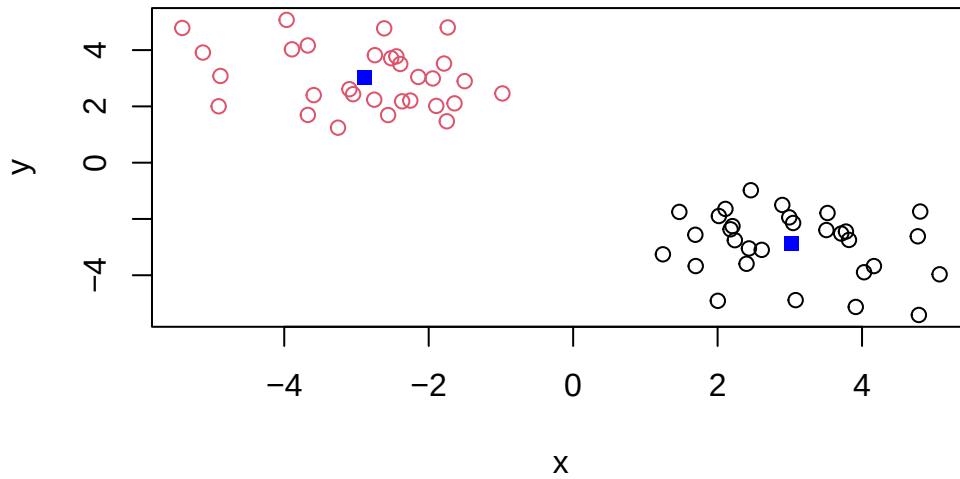
Q. What component of your result object details cluster center?

```
km$centers
```

```
      x      y
1 3.021856 -2.885675
2 -2.885675  3.021856
```

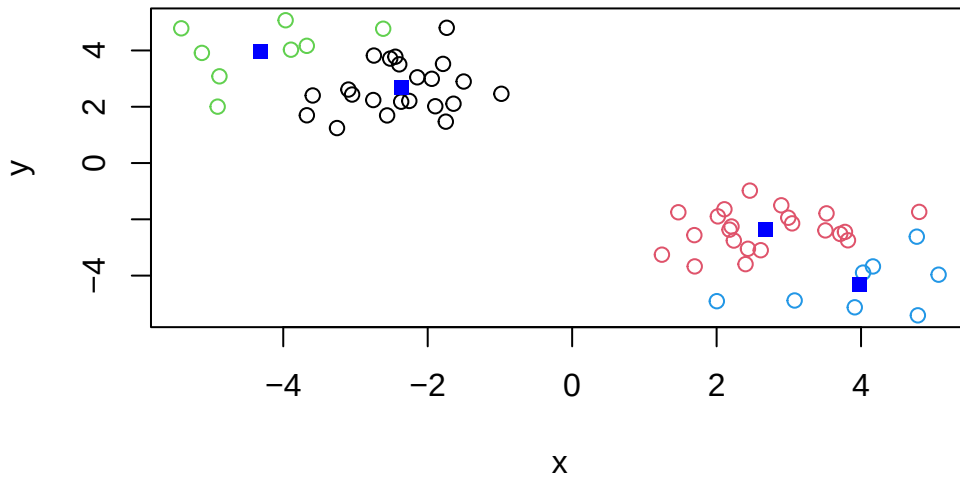
Q. Plot `z` colored by the `kmeans` cluster assignment and add cluster centers as blue points.

```
plot(z, col=c(km$cluster))
points(km$centers, col='blue', pch=15)
```



Q. Run a Kmeans clustering and plot the results asking for 4 clusters.

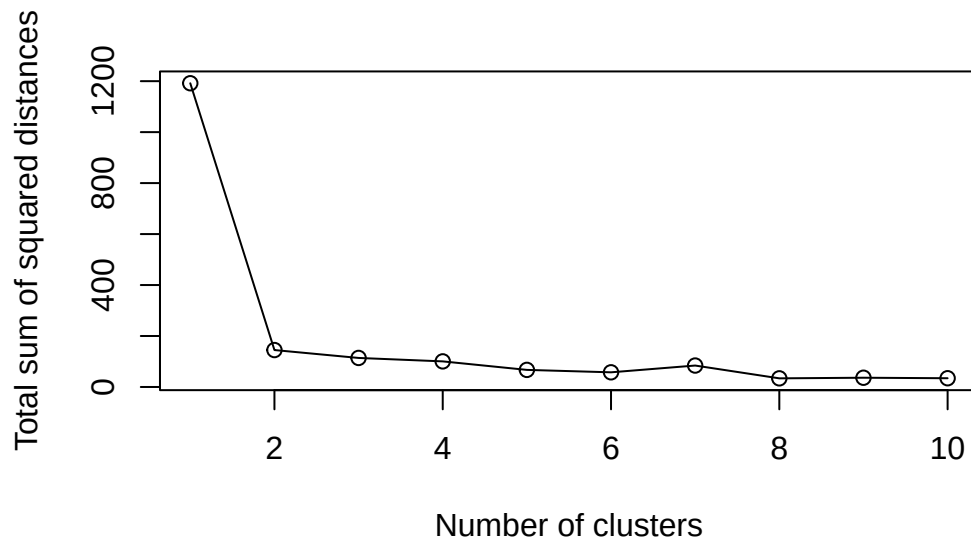
```
km4 <- kmeans(z, 4)
plot(z, col=c(km4$cluster))
points(km4$centers, col='blue', pch=15)
```



N.B. You need to tell K-means the number of clusters (i.e. set `centers=2`)!!

One approach is to try different values for `centers` and then pick the best...

```
ans <- NULL
for(i in 1:10) {
  km <- kmeans(z, i)
  ans <- c(ans, km$tot.withinss)
}
plot(ans, typ="o",
      xlab="Number of clusters",
      ylab="Total sum of squared distances")
```



Hierarchical Clustering

The main function in “base” R for Hierarchical Clustering is called `hclust()`

This function does not take your “raw” data for clustering. You must first build a “distance matrix” from your data and pass this as input to `hclust()`

```
d <- dist(z)
hc <- hclust(d)
hc
```

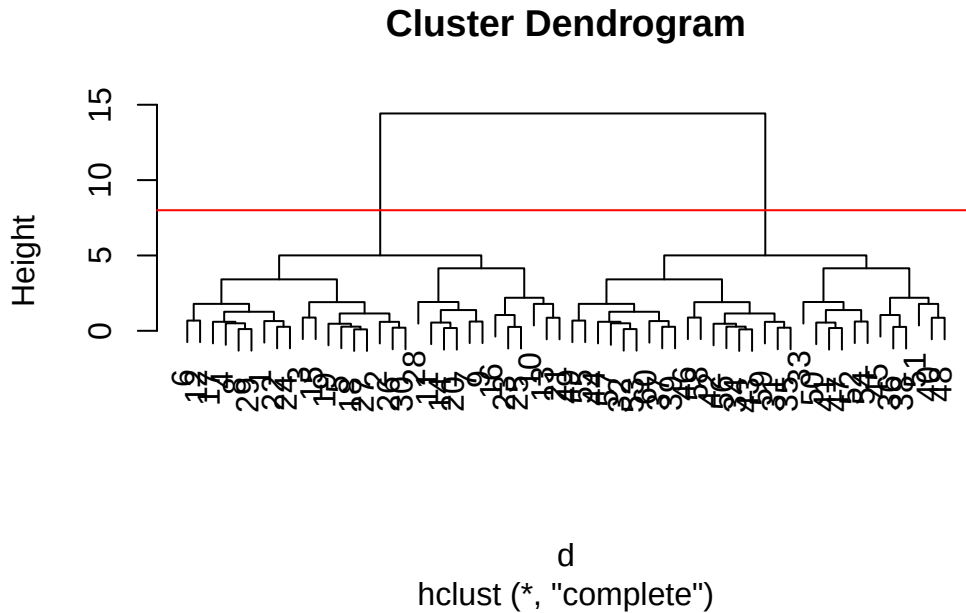
Call:

```
hclust(d = d)
```

```
Cluster method : complete
Distance       : euclidean
Number of objects: 60
```

There is a bespoke `plot()` method for `hclust()` result objects.

```
plot(hc)
abline(h=8, col="red")
```



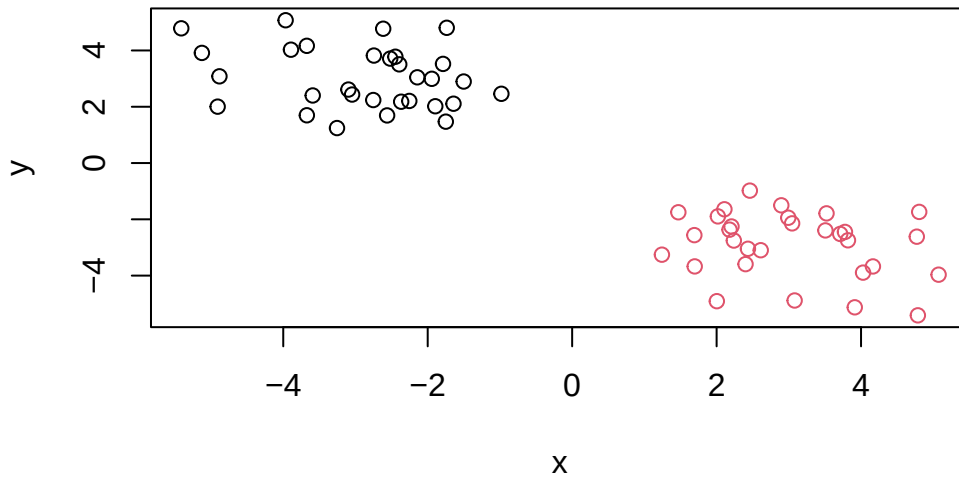
Once we have our `hclust` object (tree of cluster dendrogram), we can “*cut*” the tree to reveal the clustering pattern.

```
cutree(hc, h=4)
```

```
[1] 1 1 1 1 1 1 2 1 2 3 2 1 3 2 1 3 1 1 1 2 3 1 3 1 3 1 1 2 1 1 4 4 5 4 4 6 4 6
[39] 4 6 5 4 4 4 6 4 5 6 4 5 6 5 4 5 4 4 4 4 4 4 4 4
```

Q. Make a plot of `z` with your `hclust` results (i.e. colored by cluster membership).

```
grps <- cutree(hc, k=2)
plot(z, col=c(grps))
```



Principle Component Analysis (PCA)

PCA is a dimensionality reduction method that is popular for revealing pattern in complex datasets.

Analysis of UK food data

Let's look at some data on the eating habits of folks from the UK to see if there are pattern and trends that have some regions being distinct from others.

Data import

The data is made available in CSV format

```
url <- "https://tinyurl.com/UK-foods"
x <- read.csv(url)
```

Q1. How many rows and columns are in your new data frame named x? What R functions could you use to answer this questions?


```
dim(x)
```

```
[1] 17  5
```

Tidy the data

Fix anything that went wrong with data import

```
x <- read.csv(url, row.names=1)
head(x)
```

	England	Wales	Scotland	N.Ireland
Cheese	105	103	103	66
Carcass_meat	245	227	242	267
Other_meat	685	803	750	586
Fish	147	160	122	93
Fats_and_oils	193	235	184	209
Sugars	156	175	147	139

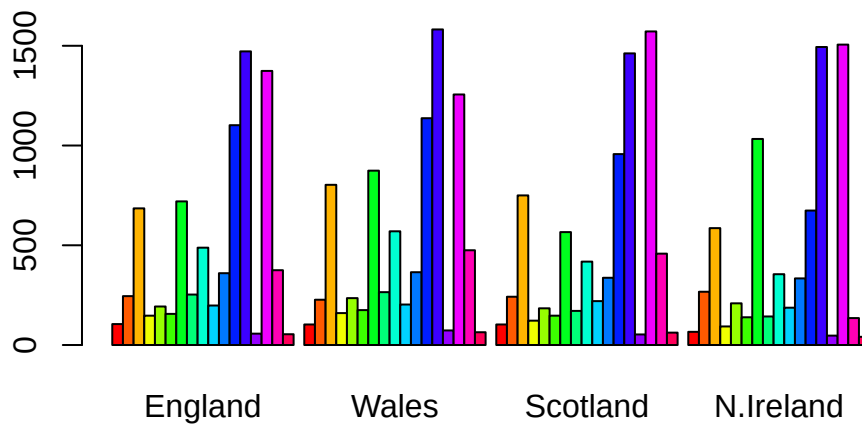
Q2. Which approach to solving the ‘row-names problem’ mentioned above do you prefer and why? Is one approach more robust than another under certain circumstances?

I prefer using `row.names=1` because `x <- x[,-1]` will cause problems if you run multiple times.

Exploratory analysis

Make some plots to help make sense of obvious trends...

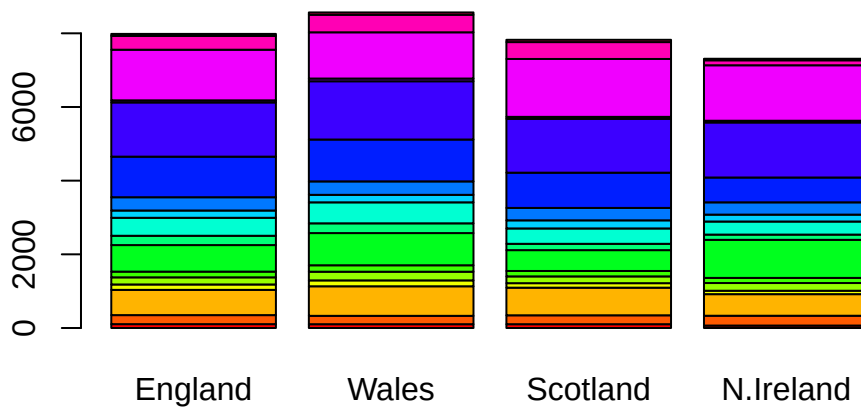
```
barplot(as.matrix(x), beside=T, col=rainbow(nrow(x)))
```



Q3: Changing what optional argument in the above `barplot()` function results in the following plot?

Changing `beside=FALSE`

```
barplot(as.matrix(x), beside=F, col=rainbow(nrow(x)))
```

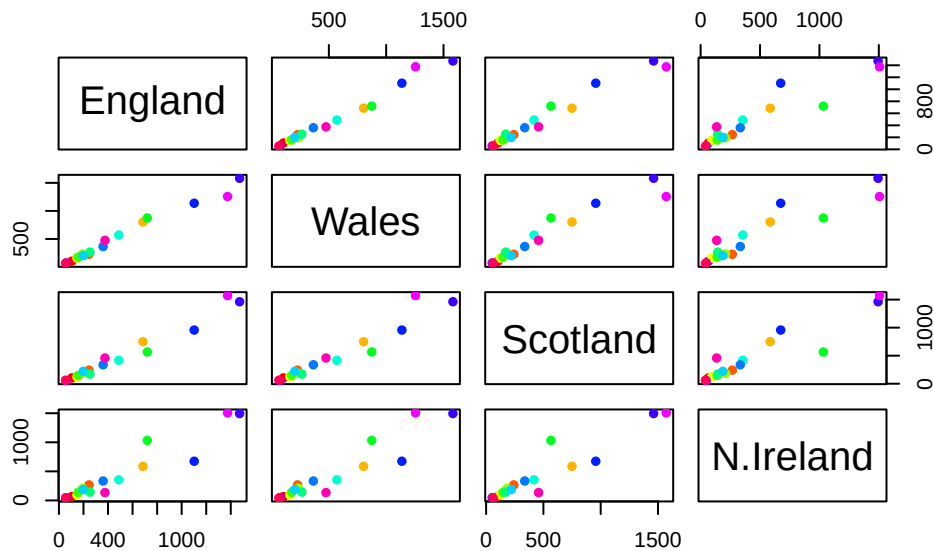


Q4: Changing what optional argument in the above `ggplot()` code results in a stacked barplot figure?

We can use `geom_col()` function to create a stacked bar chart where values are accumulated vertically

Q5: We can use the `pairs()` function to generate all pairwise plots for our countries. Can you make sense of the following code and resulting figure? What does it mean if a given point lies on the diagonal for a given plot?

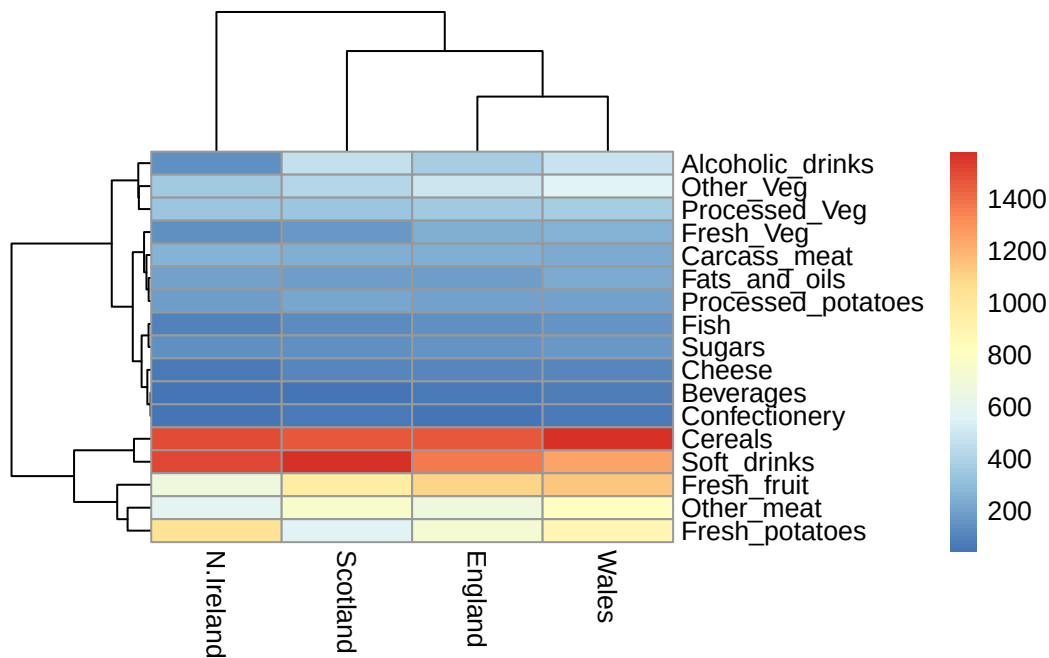
```
pairs(x, col=rainbow(nrow(x)), pch=16)
```



The `pairs()` function creates a scatterplot matrix, showing all pairwise relationships between the variables. Each small panel is a scatterplot comparing one country (x-axis) vs another (y-axis). If a given point lies on the diagonal for a given plot, it means that the two variables have equal values for that observation

```
library(pheatmap)

pheatmap( as.matrix(x) )
```



Q6. Based on the pairs and heatmap figures, which countries cluster together and what does this suggest about their food consumption patterns? Can you easily tell what the main differences between N. Ireland and the other countries of the UK in terms of this data-set?

England and Wales cluster most closely together, with Scotland joining them next, while Northern Ireland is the most distinct. This suggests England and Wales have very similar food consumption patterns, Scotland is somewhat similar, and Northern Ireland differs more. But the exact main differences aren't immediately obvious without deeper analysis.

PCA

The main function in “base” R for PCA is called `prcomp()`. This function expects the observations to be rows and the variables to be columns.

So here we need to take transpose of our `x` input object

```
pca<- prcomp(t(x))
summary(pca)
```

Importance of components:

	PC1	PC2	PC3	PC4
Standard deviation	324.1502	212.7478	73.87622	2.921e-14

```
Proportion of Variance  0.6744  0.2905  0.03503 0.000e+00
Cumulative Proportion   0.6744  0.9650  1.00000 1.000e+00
```

The returned `pca` object has components that we can use to make our main result figures:

```
attributes(pca)
```

```
$names
[1] "sdev"      "rotation" "center"    "scale"     "x"
```

```
$class
[1] "prcomp"
```

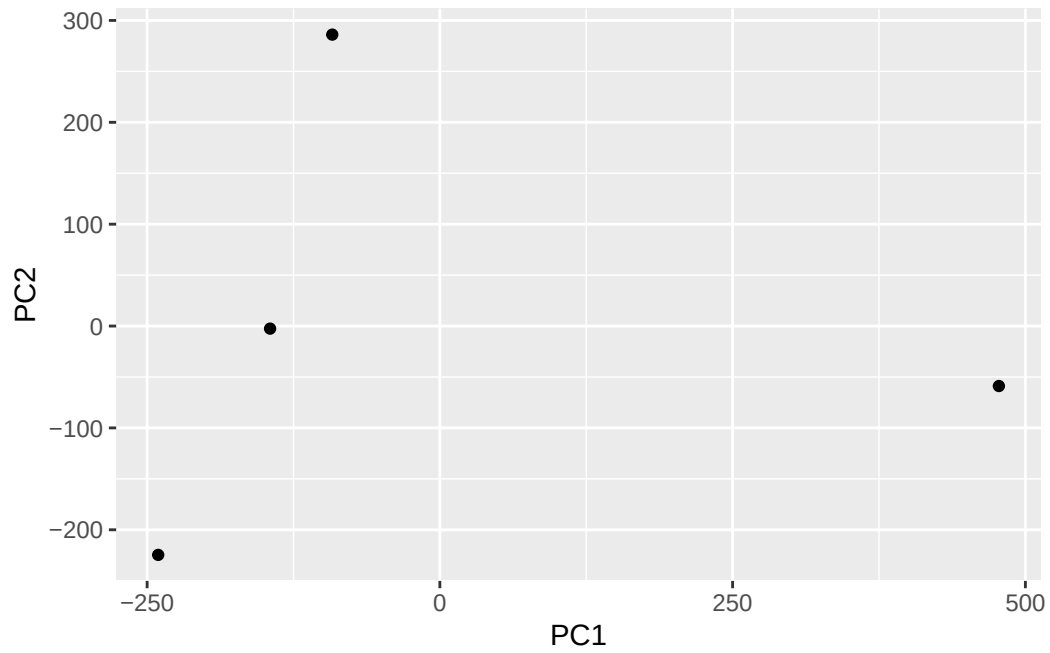
The main result figure from this analysis is called a “PC score plot” or “ordination plot” “PC plot” or “PC1 vs PC2 plot”.

This plot shows how samples (in this case countries) relate to each other along our new PC axis.

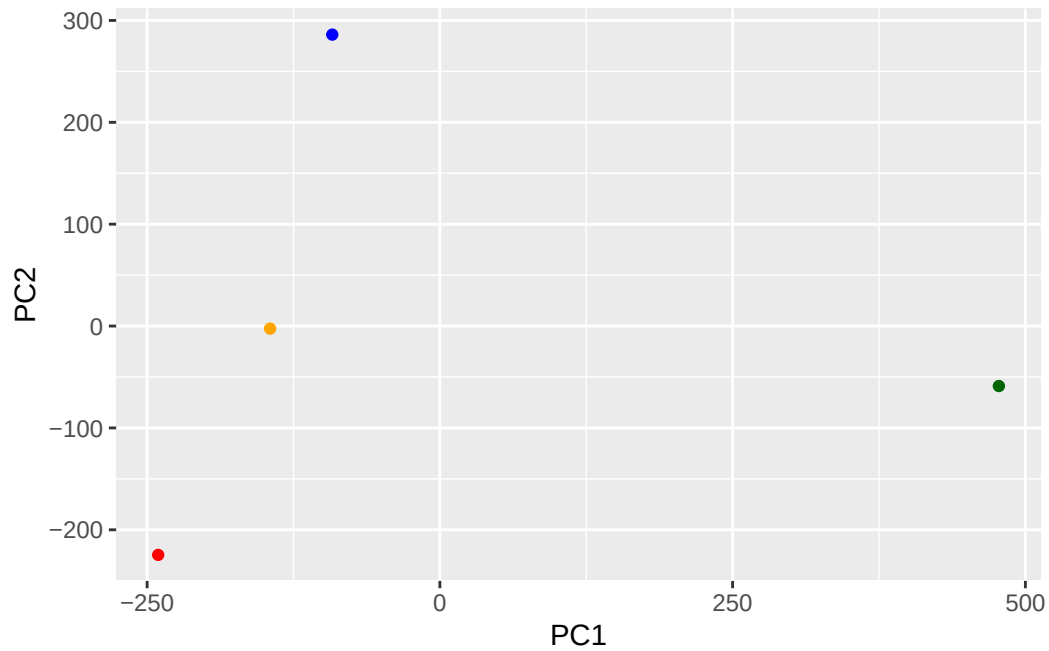
This is our new “reduced-dimensional space”. In this case 2 dimensions, PC1 and PC2, that capture most of the variance in the dataset.

Q7. Complete the code below to generate a plot of PC1 vs PC2. The second line adds text labels over the data points.

```
library(ggplot2)
ggplot(pca$x) +
  aes(PC1, PC2) +
  geom_point()
```

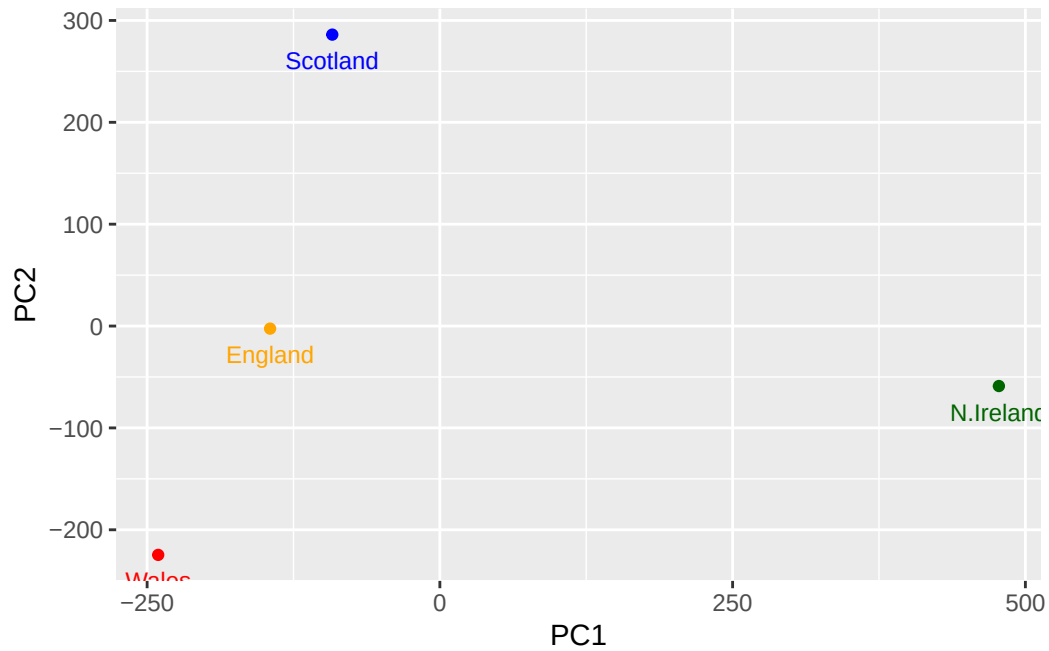


```
mycols <- c("orange", "red", "blue", "darkgreen")  
  
ggplot(pca$x) +  
  aes(PC1, PC2) +  
  geom_point(col=mycols)
```



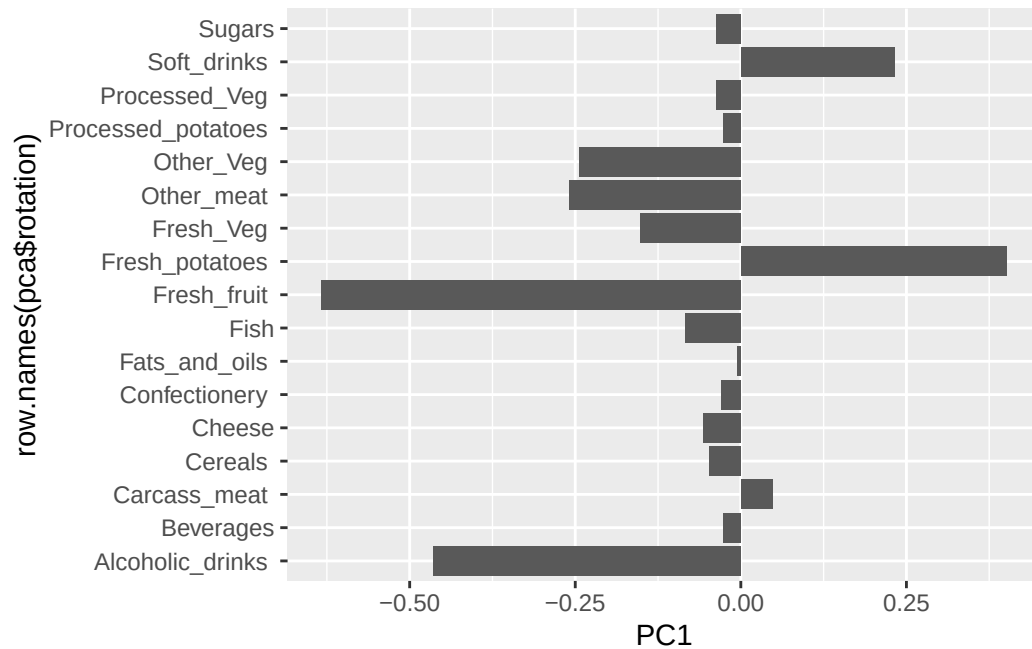
Q8. Customize your plot so that the colors of the country names match the colors in our UK and Ireland map and table at start of this document.

```
ggplot(pca$x) +  
  aes(PC1, PC2, label=row.names(pca$x)) +  
  geom_point(col=mycols) +  
  geom_text(size=3, vjust=2, col=mycols)
```

Q9: Generate a similar 'loadings plot' for PC2. What two food groups feature prominently and what does PC2 mainly tell us about?

```
ggplot(pca$rotation) +  
  aes(PC1, row.names(pca$rotation)) +  
  geom_col()
```



```
ggplot(pca$rotation) +
  aes(x = PC1,
      y = reorder(rownames(pca$rotation), PC1)) +
  geom_col(fill = "steelblue") +
  xlab("PC1 Loading Score") +
  ylab("") +
  theme_bw() +
  theme(axis.text.y = element_text(size = 9))
```

